

Early Type Ia Light Curves in Double-Degenerate Binaries

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ABSTRACT

Supernovae in binary star systems involve a hydrodynamical interaction between the ejecta and a binary companion. This collision results in shock heating and a modified density structure for the ejecta, both of which affect the light curve. As highlighted by Kasen, these considerations are particularly relevant for type Ia supernovae, as the companion is expected to be Roche-lobe filling at the time of the explosion. We simulate here the interaction between type Ia supernova ejecta and a white dwarf donor using Athena++, finding the formation of a low-density wake extending to higher velocities than the unperturbed ejecta. Radiation hydrodynamics is then used to generate synthetic light curves for the first day after the explosion for a range of viewing angles. We find that the hot, high-velocity, shocked ejecta yields $L > 10^{40}$ ergs/s over half the sky in the first few hours. The photosphere within the shock-heated ejecta cools and recedes in velocity space, partially obscuring it from view, as heating from radioactive nickel becomes increasingly important in driving the supernova's luminosity. By one day after the explosion, the luminosity measured by observers looking directly into the wake is dimmer than that of a normal type Ia supernova by 15 percent due to the modified density structure.

Keywords: Radiative transfer(1335) — Type Ia supernovae(1728) — Transient sources(1851) — Hydrodynamics(1963)

1. INTRODUCTION

Type Ia supernovae (SNIa) can be triggered by accretion onto a white dwarf (WD) from a binary companion or from the merger of two white dwarfs. In the former case, the donor may be either degenerate or non-degenerate (for a review see [Ruiter & Seitenzahl 2025](#)). Regardless, the collision between the ejecta and donor may have lasting effects both on the subsequent evolution of the donor and on the morphology of the ejecta. The effects on the donor—such as mass stripping, pollution, and entropy deposition—have been explored in a variety of hydrodynamical simulations ([Marrone et al. 2000](#); [Liu et al. 2015b](#); [Papish et al. 2015](#); [Tanikawa et al. 2018](#); [Bauer et al. 2019](#); [Wong et al. 2024](#)). Though these studies have considered both the single- and double-degenerate scenarios, the fact that

mass transfer requires the donor to fill its Roche lobe means that the solid angle subtended by the donor relative to the accretor is independent of the degeneracy of the donor, and approaches 10% of the whole sky. The encounter of the expanding ejecta with the donor results in a wake that is nearly a factor of two larger.

In the double-degenerate case, the supernova may be achieved by the detonation of a thin shell of accreted helium surrounding a carbon-oxygen (CO) WD ([Boos et al. 2021](#)), driving a converging shock wave into the accretor which then triggers a second detonation ([Shen & Bildsten 2014](#)). This formation channel necessitates a small orbital separation at the time of detonation, with an orbital velocity of ≈ 800 – 1500 km/s for the donor. The population of hypervelocity white dwarfs observed by Gaia ([Shen et al. 2018](#); [El-Badry et al. 2023](#)) and SDSS ([Hollands et al. 2025](#)) are likely to be the surviving donors of double-degenerate SNIa. Furthermore, the deposition of entropy into the interior of the donor places it in an unusual location on the Hertzsprung-Russell di-

59 agram (Bauer et al. 2019; Wong et al. 2024; Zhang et al.
60 2019).

61 Our focus here is on the thermodynamics and sub-
62 sequent radiation from the conical wake in the ejecta
63 created by the donor interaction. Filled with low den-
64 sity, high temperature gas due to heating from the bow
65 shock adorning the donor, this wake is a permanent fea-
66 ture of the ejecta. Kasen (2010) showed that the wake
67 has observational consequences for the early emission
68 from the supernova, which has been substantiated in
69 several cases: Marion et al. (2016) concluded that a blue
70 bump in SN 2012cg was due to interaction with a non-
71 degenerate donor, Ni et al. (2022) observed a plateau in
72 the B-band between 1.0 and 12.4 hours after first light
73 in SN 2018aoz, and Liu et al. (2015a) showed that a UV
74 flash in SN Ia iPTF14atg was due to interaction with a
75 non-degenerate donor. On the other hand, Piro & Mo-
76 rozova (2016) and Noebauer et al. (2017) both note that
77 an excess of ^{56}Ni in the outermost ejecta can masquer-
78 ade as a collision with the circumstellar medium (CSM)
79 or a donor, and Hu et al. (2023) show that some (but
80 not all) SNIa multi-band light curves can be explained
81 by CSM interaction.

82 Kasen (2010) performed the first calculation for the
83 single-degenerate case. Multiple groups have since stud-
84 ied the light curves and spectra of double-degenerate
85 SNIa in various scenarios (Shen et al. 2021; Pollin et al.
86 2024; Boos et al. 2024) but focused on $t \gtrsim 5$ days af-
87 ter shock breakout, so it remains unclear to what de-
88 gree this affects the synthetic observables at $t \leq 1$ day.
89 The double-degenerate case requires a special consider-
90 ation of the equation of state of the ejecta. It is initially
91 gas pressure-dominated, but becomes radiation-pressure
92 dominated due to the bow shock. To calculate the re-
93 sulting entropy of the post-shock material, a full equa-
94 tion of state spanning both regimes must be used. To
95 this end, we perform hydrodynamical simulations of the
96 donor interaction with SNIa ejecta to obtain the result-
97 ing ejecta structure. We then homologously evolve this
98 structure and apply radiation hydrodynamics to predict
99 the supernova luminosity from all parts of the sky in the
100 first day after the explosion.

101 We organize this paper as follows. The setup for our
102 hydrodynamical simulations is described in section 2,
103 where we show the results of these calculations. Sub-
104 sequent homologous evolution of the ejecta (including
105 ^{56}Ni heating) is discussed in section 3. Radiation hy-
106 drodynamics is used to obtain bolometric light curves of
107 these events in section 4. We discuss these results and
108 explore the potential for future work in section 5.

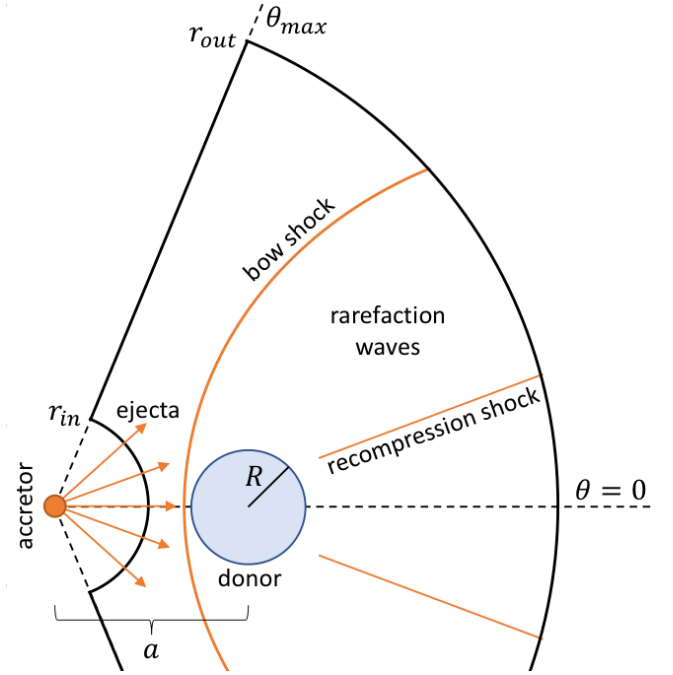


Figure 1. Diagram of our numerical setup highlighting several features of the resulting wake.

109 2. HYDRODYNAMICAL INTERACTION AND THE 110 WAKE STRUCTURE

111 We perform hydrodynamical simulations using
112 Athena++ (Stone et al. 2020), which solves the Eu-
113 ler equations

$$114 \quad \frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{v} = 0, \quad (1)$$

$$115 \quad \frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v} \mathbf{v}^T + P \mathbb{I}) = 0, \quad (2)$$

$$116 \quad \frac{\partial u}{\partial t} + \nabla \cdot (e + P) \mathbf{v} = 0 \quad (3)$$

117 for material with density ρ , fluid velocity $\mathbf{v}$, pressure P ,
118 energy density $e = u + \rho v^2/2$, and internal energy u ,
119 where $\mathbb{I}$ is the identity matrix. The Athena++ setup
120 for all simulations reported in this paper as well as
121 our analysis scripts are located at <https://github.com/Dota2Xin/earlyLightCurvesReproduce.git>.
122

123 Most of the ejecta is initially gas-pressure domi-
124 nated, but becomes radiation-pressure dominated when
125 shocked or later heated by radioactive decay. We treat
126 the gas as ideal and include radiation pressure assuming
127 local thermodynamic equilibrium, so that the internal
128 energy and pressure are

$$129 \quad u = a_r T^4 + \frac{3}{2} \frac{\rho k_B T}{\mu m_p}, \quad (4)$$

$$130 \quad P = \frac{1}{3} a_r T^4 + \frac{\rho k_B T}{\mu m_p}, \quad (5)$$

where T is the gas and radiation temperature, a_r is the radiation constant, k_B is the Boltzmann constant, m_p is the proton mass, and μ is the mean molecular weight of the fully ionized plasma which we set to $4/3$ (fully ionized He). Closure of the Euler equations requires conversion between pressure and energy, and thus a determination of T . We accomplish this via a quartic solver which performs a root find to solve equations of the form $T^4 + BT - A = 0$. We also require the adiabatic sound speed $c_s^2 = \Gamma_1 P / \rho$. Here Γ_1 is given by

$$\Gamma_1 = \left(\frac{\partial P}{\partial \rho} \right)_S = \frac{32 - 24\beta - 3\beta^2}{24 - 21\beta} \quad (6)$$

(Mihalas & Mihalas 1984), where β is the ratio of gas pressure to total pressure.

We use a 3-D spherical-polar mesh with the explosion at the origin. The domain extends from an inner radial boundary $r_{\text{in}} = 0.13 R_\odot$ —where we inject the SN ejecta—to an outer boundary at $r_{\text{out}} = 10.8 R_\odot$ where the gas freely outflows (Fig. 1). This r_{out} is sufficiently large to ensure that the gas exiting this boundary has reached homology, which we demonstrate below. The donor is located on the $\theta = 0$ pole at an orbital separation of $a = 0.27 R_\odot$ with radius $R = 0.08 R_\odot$ and mass $M_d = 0.344 M_\odot$, which is identical to Model 1 of Bauer et al. (2019). The domain extends out to a maximum polar angle of 80° , while the azimuthal angle ϕ spans the full range of 0 to 2π . The donor is treated as a reflective spherical boundary. We use the Gaussian ejecta profile from Wong et al. (2024) with explosion energy $E_{\text{ej}} = 0.97 \times 10^{51}$ ergs and ejecta mass $M_{\text{ej}} = 0.89 M_\odot$. More detailed information on the numerical setup is available in Prust et al. (2025).

We simulate the collision between the ejecta and the donor using the setup discussed above, terminating at $t = 1000$ s. Slice plots of our results at $t = 200$ s are shown in Fig. 2, wherein we see a bow shock adorning the donor. A weaker recompression shock also forms as ejecta converges on the $\theta = 0$ pole downstream of the donor. These features lead to the formation of a low-density, high-temperature wake in the ejecta.

In Fig. 3 we show several properties of the ejecta as it outflows from the domain at r_{out} . The density (upper left panel) shows discontinuities at both shocks, with the density dropping dramatically near the pole. This differs from Kasen (2010), where the recompression shock is absent and the density is uniform for small θ . The low density within the wake affects the light curve, as we show in section 4. The velocity in the θ direction reveals the presence of rarefaction waves between the shocks (upper right). We see that the temperature (lower left) is at least an order of magnitude greater for

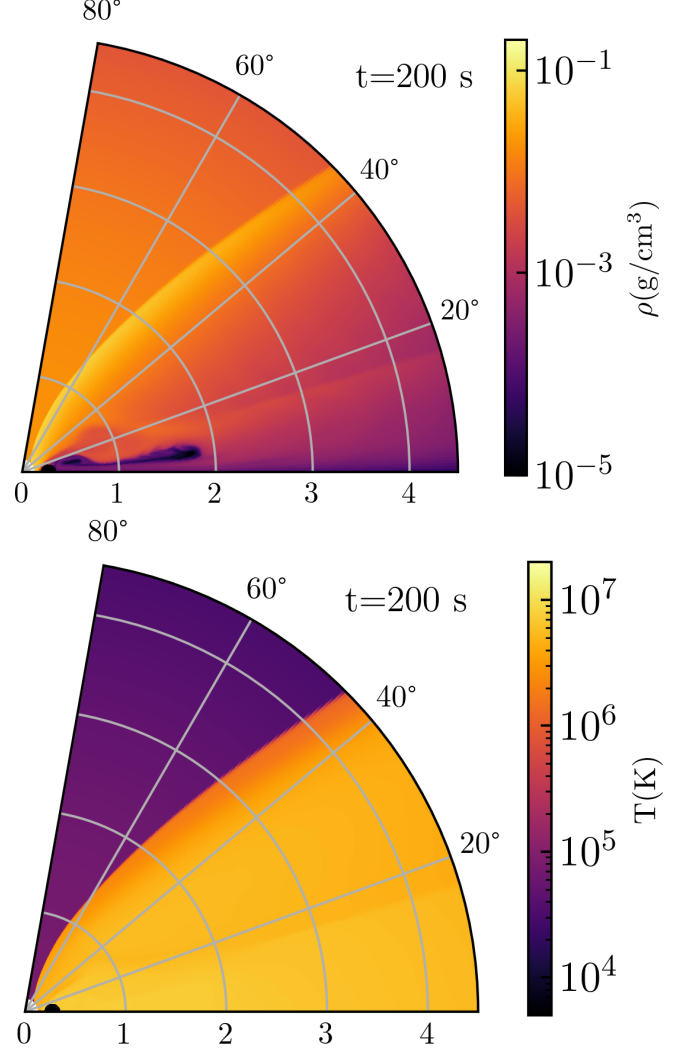


Figure 2. Slice plots showing density (top) and temperature (bottom) on the $\phi = 0$ plane 200 seconds after the supernova.

the shocked ejecta. The widening of the shock cone is also apparent in this plot, encompassing $\theta \lesssim 50^\circ$ by the end of the run. This means that roughly 18% of the ejecta (by solid angle) has experienced shock heating. Furthermore, whereas the unshocked ejecta remains gas pressure-dominated, the shocked ejecta is primarily radiation pressure-dominated (lower right).

The use of a rigid boundary to model the donor allows us to directly compute the momentum transferred to the donor in the collision by simply integrating the pressure over the donor surface. Here the pressure is computed during the Riemann solve and is taken to be the momentum flux $\rho \mathbf{v} \mathbf{v}^T + P \mathbb{I}$ normal to the donor surface. We measure the total momentum imparted to the donor as $\Delta p = 96 M_\odot \text{km/s}$, which exceeds the idealized estimate by $\approx 20\%$ (see Appendix A). Though our

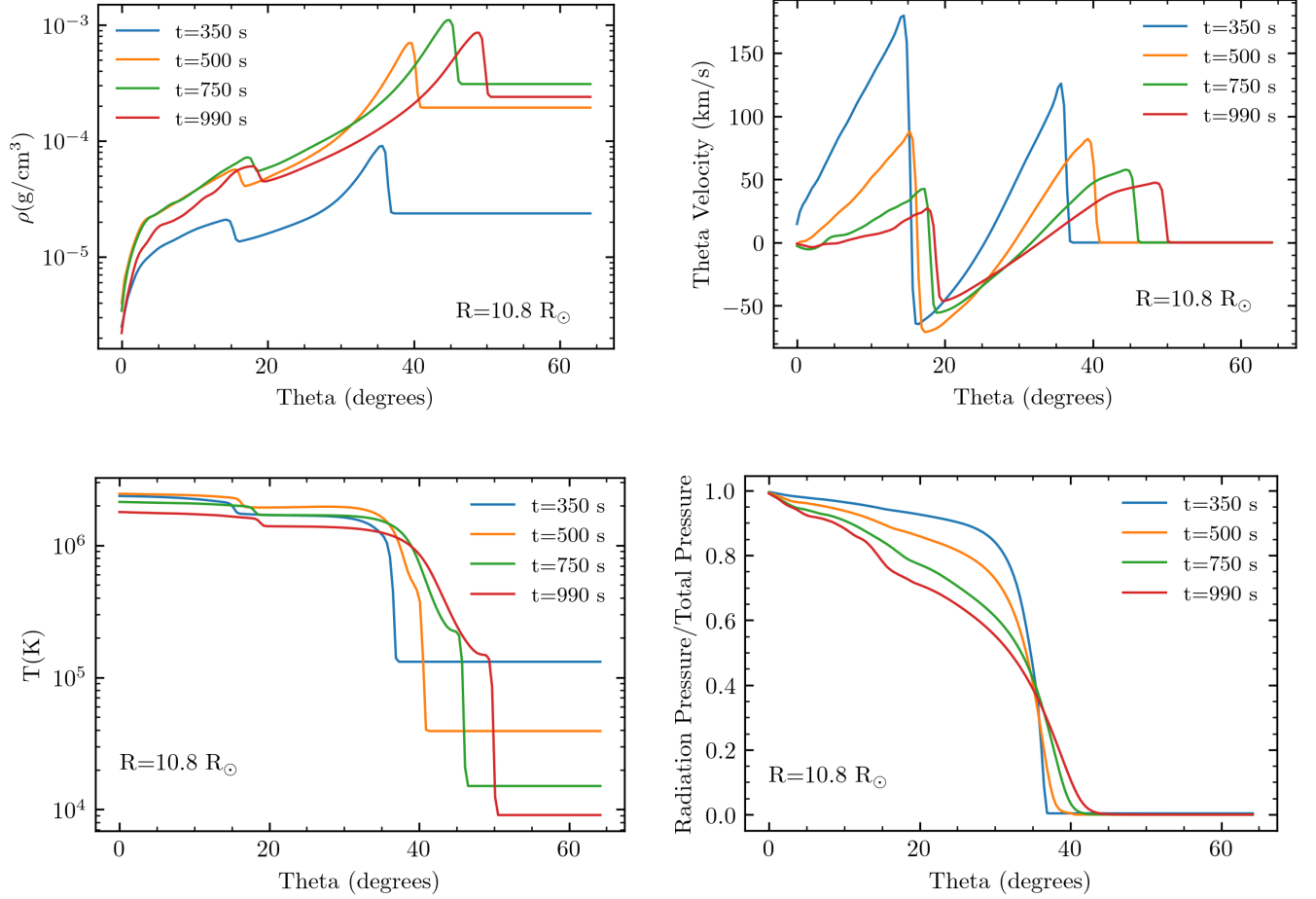


Figure 3. Several properties of the ejecta measured at $r_{\text{out}} = 10.8 R_{\odot}$ plotted vs θ at various times. The density (upper left) shows a rapid dropoff at small θ , differing from Kasen (2010). The transverse velocity (upper right) illustrates the positions of the shocks and shows rarefaction waves between them. The temperature (lower left) reveals that the shock-heated ejecta is much hotter than the unshocked gas, causing much of the shocked ejecta to be radiation pressure-dominated (lower right).

donor is fixed to the grid, the kick velocity associated with this impulse would be $\Delta p/M_d = 279$ km/s, which is nearly perpendicular to the orbital speed $v_{\text{orb}} = 691$ km/s. This exceeds the kick velocities found in [Bauer et al. \(2019\)](#) and [Wong et al. \(2024\)](#) for comparable progenitors, both of which allowed for mass loss from the donor. The unbound mass carried away a portion of the imparted momentum which may otherwise have contributed to the kick velocity, suggesting that the treatment of the donor as a fixed, rigid sphere gives an upper limit on the imparted momentum.

3. EVOLVING OPTICALLY THICK EJECTA

Because the vast majority of the ejecta is optically thick for the first few hours following the supernova, it is undesirable to conduct a full 3-D radiation hydrodynamical simulation for this period. Instead, we evolve the ejecta forward in time assuming homology until the outer layers begin to radiate. Here we consider only the material which exited the outer boundary of the simulation discussed in the previous section, as the gas remaining inside the domain remains far inside the photosphere for the first day.

We first confirm that the gas which passed through r_{out} is in homology by checking the ratio of gas plus radiation pressure to ram pressure, finding that $P/(\rho v^2/2) \ll 1$. We also check that the flow can be approximated as radial: the upper right panel of [Fig. 3](#) shows that the transverse velocity is ≈ 100 km/s, which is two orders of magnitude less than the radial velocity. Thus, the density of each Lagrangian parcel subsequently evolves as $\rho \propto t^{-3}$. The possible presence of ^{56}Ni in the ejecta requires including the effects of radioactive heating, at a rate

$$\epsilon = \frac{X_{56} E_{\text{nuc}}}{A m_p \tau_{56}} e^{-t/\tau_{56}}, \quad (7)$$

where $X_{56} = 0.04$ is a uniform mass fraction of ^{56}Ni placed in our ejecta, $E_{\text{nuc}} = 1.72$ MeV is the energy released in the $^{56}\text{Ni} \rightarrow ^{56}\text{Co}$ decay, $\tau_{56} = 757,728$ s is the e-folding time of ^{56}Ni decay, and $A = 56$ is its mass number. The temperature of each parcel then evolves as

$$T \frac{ds}{dt} = \frac{X_{56} E_{\text{nuc}}}{A m_p \tau_{56}} e^{-t/\tau_{56}}, \quad (8)$$

where s is the specific entropy (including radiation):

$$s = \frac{k_B}{\mu m_p} \ln \left(\frac{T^{3/2}}{\rho} \right) + \frac{4a_r T^3}{3\rho}. \quad (9)$$

Equations (8) and (9) are then solved for $T(t)$.

Our numerical scheme for evolving the fluid state of each parcel is as follows. We start at time $t_0 = 1000$ s with $s = s_0$, $T = T_0$, and $\rho = \rho_0$ and evolve to $t = 7200$ s. We divide this time interval into 10^4 equal time steps Δt , which is sufficient for convergence. For each time step, we evolve the parcel adiabatically and then update its entropy:

1. Evolve the density as $\rho_{i+1} = \rho_0 (t_0/t_{i+1})^3$.
2. Adiabatically evolve the gas to t_{i+1} from t_i by root finding (via the bisection method) for T_{i+1} in (9) with $\rho = \rho_{i+1}$ and $s = s_i$.
3. Update the entropy according to (8):

$$s_{i+1} = s_i + \Delta t \left(\frac{X E_{\text{nuc}}}{A m_p \tau_{56}} \frac{2}{T_{i+1} + T_i} \right) e^{-t_i/\tau_{56}} \quad (10)$$

This evolution yields a distribution of Lagrangian parcels, which we map back onto a spherical-polar grid to perform the radiation hydrodynamical simulation. Here each grid cell adopted the fluid state of the Lagrangian parcel nearest to its cell center. The results of this effort are shown in [Fig. 4](#), showing a clear distinction between unshocked ejecta, ejecta which was shocked by the bow shock, and ejecta which was shocked twice. Notably, the shocked ejecta protrudes far beyond unshocked ejecta due to the supersonic expansion on the downstream side of the donor, which accelerated the ejecta by as much as $\approx 50\%$ relative to the unshocked ejecta. As we will see in the following section, this means that the shocked ejecta is visible from a broad range of viewing angles.

4. RADIATION AND LIGHT CURVE

To analyze the radiation emitted by the ejecta we perform radiation hydrodynamics simulations using [Athena++ \(Jiang 2021\)](#), which solves the fluid equations coupled to radiation source terms S_r . The time-dependent radiative transfer equation is solved iteratively, ensuring an accurate solution in both the diffusive and free-streaming limits. Additionally, the implicit solver described in [Jiang \(2021\)](#) removes the speed of light as a timestep constraint, greatly reducing the computational cost. The relevant equations are:

$$\frac{\partial I}{\partial t} + \mathbf{c} \mathbf{n} \cdot \nabla \mathbf{I} = c(\eta - \chi I), \quad (11)$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot \rho \mathbf{v} \mathbf{v} + \nabla P = -\mathbf{S}_r(\mathbf{P}), \quad (12)$$

$$\frac{\partial u_g}{\partial t} + \nabla \cdot (u_g + P) \mathbf{v} = -S_r(u_g), \quad (13)$$

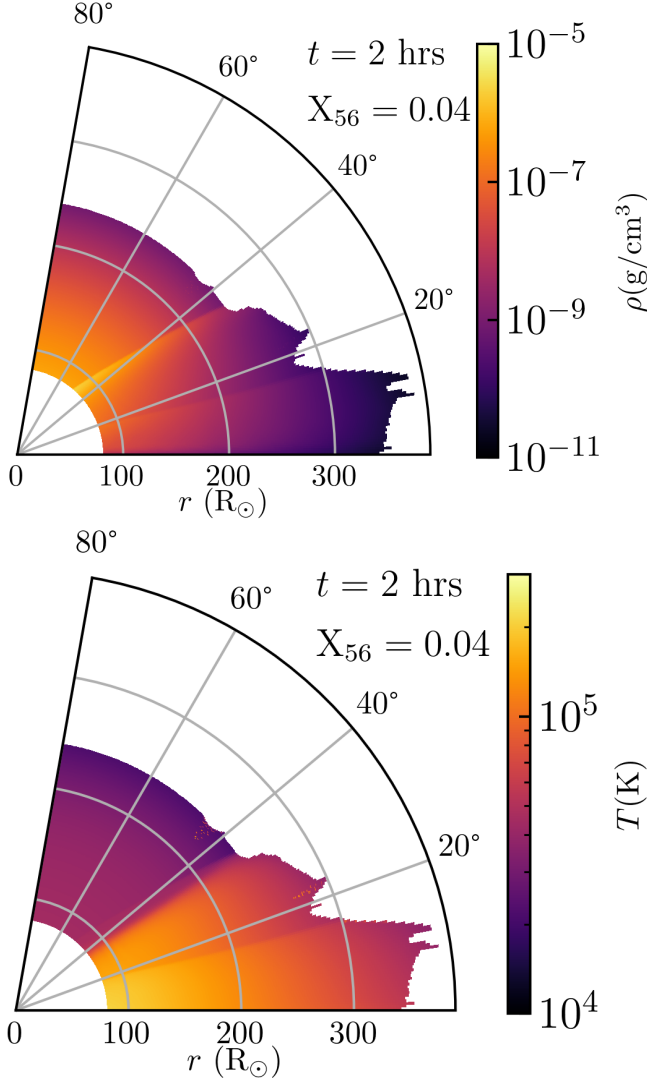


Figure 4. Slice plots showing density (top) and temperature (bottom) on the $\phi = 0$ plane after evolving the material leaving the edge of the original simulation to $t = 2$ hours after the explosion for $X_{56} = 0.04$.

$$S_I = \rho \Lambda^{-3} \left[(\kappa_s + \kappa_a)(J_0 - I_0) + (\kappa_a + \kappa_{\delta P}) \left(\frac{a_r T^4}{4\pi} - J_0 \right) \right], \quad (14)$$

$$S_r(E) = 4\pi c \int S_I d\Omega, \quad (15)$$

$$\mathbf{S}_r(\mathbf{P}) = 4\pi \int \mathbf{n} S_I d\Omega, \quad (16)$$

$$\Lambda(\mathbf{n}, \mathbf{v}) = \gamma(1 - \mathbf{n} \cdot \mathbf{v}/c). \quad (17)$$

Here c is the speed of light, η is the lab frame emissivity, χ is the lab frame opacity, u_g is the internal energy of

the gas, and I is the frequency-integrated specific intensity of the radiation field which is a function of space, time, and angular direction $\mathbf{n}$. In the source terms γ is the Lorentz factor, κ_s is the scattering opacity, κ_a is the Rosseland mean absorption opacity, and $\kappa_{\delta P}$ is the difference between the Planck mean and Rosseland mean opacities. The variables I_0 and J_0 are the radiation specific intensity and flux in the comoving frame of the ejecta, respectively. For the opacity, we want $\kappa_s + \kappa_a$ (absorbing + scattering) to be equal to $0.2 \text{ cm}^2/\text{g}$ for Thompson scattering in the absence of hydrogen. We choose $\kappa_s = 0.17 \text{ cm}^2/\text{g}$ and $\kappa_a = 0.03 \text{ cm}^2/\text{g}$. The artificially high absorption scattering ensures that gas and radiation are strongly coupled in our simulation while ensuring that the total opacity is $0.2 \text{ cm}^2/\text{g}$. We implement radioactive heating from the ^{56}Ni decay chain by adding a local energy source term.

As kinetic energy accounts for the vast majority of the explosion energy, numerical errors during inversions between the gas pressure and the gas total energy can cause erroneous conversions of kinetic to thermal energy. This excess heating then leaks into the radiation field and causes errors. To combat this, we use a high spatial resolution on a small domain spanning $r = 172.6 R_\odot$ to $r = 1,649.8 R_\odot$. At $t = 8$ hrs we then re-map the ejecta and the radiation field onto a larger grid spanning $r = 710.7 R_\odot$ to $r = 4,822.4 R_\odot$ allowing the integration to continue to a final time of $t_f = 23.9$ hrs. For both of these simulations a radial scaling factor of $x = 1.0012$ was used to construct the mesh.

We initialize our radiation hydrodynamics simulations by depositing the Lagrangian fluid elements described in the previous section onto a spherical-polar grid via a nearest neighbor search. The simulation now spans the full range of θ and ϕ with the supernova at the origin so as to generate synthetic light curves for observers at all angles. The radiation field is additionally specified using 80 angles at every point whose discretization scheme is described in Jiang (2021). For this run the domain contains $n_r \times n_\theta \times n_\phi = 2304 \times 300 \times 30 = 20,736,000$ cells.

We set the ambient medium to have density $\rho = 2.42 \times 10^{-22} \text{ g/cm}^3$ and pressure $P = 4.13 \times 10^{-17} \text{ erg/cm}^3$. The low pressure ensures that the ambient medium generates a negligible amount of radiation. When the ejecta collides with the ambient medium it creates an energy flux of $\rho v^3/2$ which leads to a luminosity of $\sim \rho v^3 r^2$, which is negligible given our choice of density.

Boundary conditions were set to allow radiation and fluid to freely exit through the outer radial boundary. At the inner radial boundary we insert material with the same properties (ρ , v_r , etc...) as the material already at

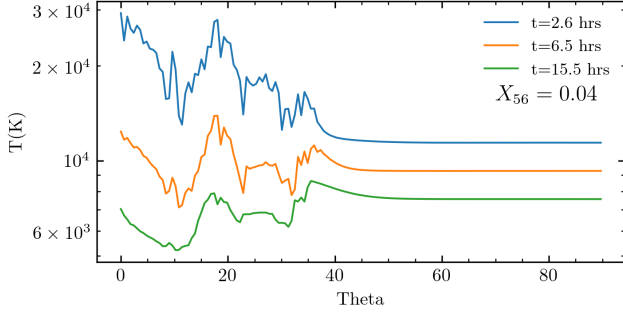


Figure 5. Temperature as a function of θ at $\tau = 1$ (integrated radially) for a set of times after the explosion.

the inner boundary. This ensures that the fluid that was initialized into the simulation does not experience abnormal pressure gradients and flows outwards as expected. This material remains optically thick and does not affect the light curve during the first day.

4.1. Simulation Results

We first consider the photosphere and how it evolves with time. Fig. 5 shows the temperature of the photosphere at $t = 2.6, 6.5$, and 15.5 hrs. The wake drops in temperature more quickly than the unshocked material due its lower density. The unshocked material is initially gas-pressure dominated so that the radioactive heating has a larger effect on its temperature than in the shocked material, which is radiation-pressure dominated (both due to its higher temperature and lower density). The full temperature profiles at two representative times of $t = 5.0$ and 18.5 hrs are plotted in Fig. 6 as well as isolines of radially integrated optical depth τ to illustrate the geometry and temperature of the photosphere and how it changes with time. The photosphere becomes more uniform in radius as time progresses.

To generate synthetic light curves, the intensities along all 80 angles are stored at every timestep at a measurement radius $R_m(t) = v_0 t + R_0$, where $R_0 = 431.5 R_\odot$, $v_0 = 37,000$ km/s, and t is measured with respect to the start of the simulation. This places $R_m(t)$ safely outside of the ejecta. The radiation energy density at this radius which is plotted in Fig. 7, from which it is evident that the shocked material produces more radiation than the unshocked material with the amount steadily decreasing with θ . The noise in Fig. 7 is due to the angular discretization scheme used in Athena++. Most importantly, the radiation from the shocked material extends to angles much larger than that of the wake itself due to the radial protrusion of the shocked ejecta, as is visible in Fig. 4.

4.2. Light Curves for all Observers

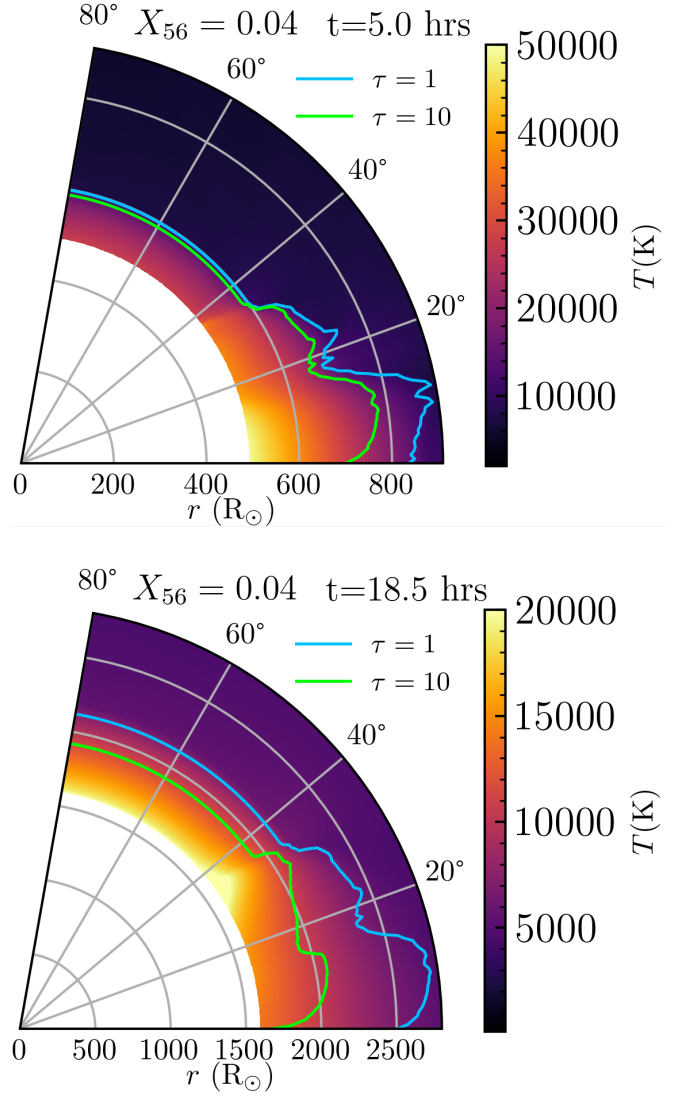


Figure 6. Plots showing temperature and the $\tau = 1$ and $\tau = 10$ optical thickness isolines on the $\phi = 0$ plane at $t = 5.0$ hrs (top) and $t = 18.5$ hrs (bottom) assuming $X_{56} = 0.04$.

We now determine the luminosity seen by observers at various angles. For an observer viewing from a direction $\hat{n}_O$, the luminosity is given by

$$L = 4\pi R^2 \int_S I(R, \theta, \phi, \hat{n}_O) (\hat{r} \cdot \hat{n}_O) \chi(\hat{r} \cdot \hat{n}_O) d\Omega, \quad (18)$$

where $I(R, \theta, \phi, \hat{n}_O)$ is the intensity of the radiation field at the point (R, θ, ϕ) radiating into the direction $\hat{n}_O$, χ is the step function, and S is the sphere where intensities were recorded.

To ensure that the simulation is self-consistent, we also check that the radiation flux F at the photosphere results in a luminosity $L = 4\pi R^2 F$ that is consistent

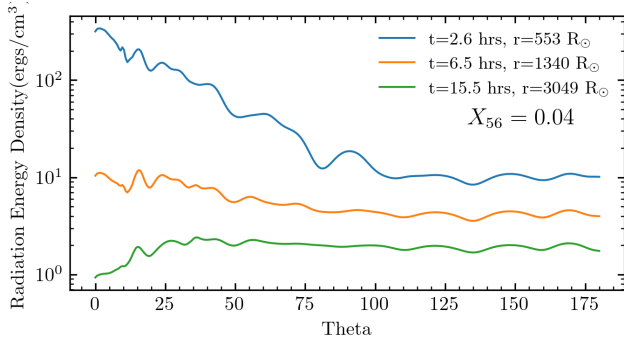


Figure 7. Radiation energy density at the measurement radius R_m vs θ at various times.

with Eq. 18. As the photosphere and measurement radius are only a few light-seconds apart, this comparison is legitimate. The two luminosities are equal to within a few percent. As a second check, we compare the co-moving flux output by Athena++ to the slope of the radiation energy density, $(d(caT^4)/d\tau)/3$, for material with $\tau > 10$. We find excellent agreement, confirming that Athena++ properly captures behavior in the diffusive limit.

Each of the angular ordinates for the radiation field is specified by a (θ_k, ϕ_k) . We sort these ordinates into 8 distinct bins in θ , where inside each bin the θ_k values deviate from one another by no more than 1.8° with each bin having a mean θ value θ_i . We calculate $L(t)$ for an observer at each of the 80 ordinates by setting $\hat{n}_O$ to the unit vector specified by (θ_k, ϕ_k) . Then, for all θ bins, we average the luminosity of the observers within them to obtain $L(t)$ for an observer at θ_i .

We show the full family of light curves from 2 to 8 hours after the explosion in Fig. 8, as well as an analytical model from Piro (2012). Observers at large θ are seeing the “back side” of the explosion, opposite to that of the wake. In the absence of a companion interaction, the early light-curve of Type Ia SNe depends on the abundances and locations of radioactive elements in the outermost ejecta (Richards et al. 2021; Magee & Maguire 2020; Magee et al. 2020, 2018; Noebauer et al. 2017; Piro & Morozova 2016; Piro & Nakar 2014, 2013; Piro 2012), as that heating maintains the high temperatures in the ejecta during homologous expansion. Piro (2012) studied the case closest to ours—a uniform ^{56}Ni distribution—making a prediction plotted with the label “Analytics” which is in reasonable agreement with our results given the differing assumptions in the ejecta density profile. For observers able to see the wake, the luminosity is initially high, specifically for an observer viewing from $\theta = 18^\circ$ it is 1.52×10^{40} ergs/s at $t = 2.9$ hrs as compared to the 3.92×10^{39} ergs/s that an observer

viewing from the backside sees. For the observers that see the wake this luminosity decays with time, in agreement with Kasen (2010). Though Kasen (2010) focused on larger orbital separations, his equation (22) predicts a luminosity of 2.6×10^{41} ergs/s at $t = 2.9$ hrs, roughly an order of magnitude larger than what we find at this time. At early times, $L(t)$ decreases with θ as the photosphere recedes and the observer sees less of the shocked material. However, the effect of the shock can still be seen at quite large angles up to $\theta = 142^\circ$. This means that while the strongest features of the effect are visible to around $\theta = 55^\circ$ (roughly 20% of the sky), the effect can still be seen to a moderate degree over more than half of the sky.

At later times, the angular contrast fades, and eventually all material approaches a rising light curve driven by radioactive heating. The light curve for $t = 2$ hrs through $t = 24$ hrs is plotted in Fig. 9. Here all viewing angles at sufficiently high θ ($\gtrsim 50^\circ$) eventually converge to nearly identical solutions. However, the observers looking directly into the wake see the luminosity dip below that of the unshocked material in an inversion of the behavior at earlier times. We find that observers at $\theta = 18^\circ$ at $t = 23$ hrs see 86% of the value seen by observers viewing the back side of the collision, and observers at $\theta = 38^\circ$ see 93% of this value. This persists for the remainder of our integration, and for angles less than $\approx 50^\circ$ the observed luminosity is persistently $\approx 15\%$ lower than from the back side. This is due to the modified density structure within the wake, which affects both the radioactive heating and the subsequent radiation transfer. Whether this persists up to the peak of the light curve is a question we leave to future work.

5. DISCUSSION & CONCLUSIONS

In this paper we explored the ejecta interaction with the donor star in a double degenerate type Ia supernova and its affect on the ejecta structure and early light curve. The shock heating leads to observers seeing the shocked material as brighter than the unshocked material for at least the first day. The ejecta morphology was also explored, finding that the nature of the shocks behind the donor leads to ejecta moving faster in the wake, getting ahead of the unshocked ejecta. This allows for observers at large angles to the wake to see the brighter emission from the shocked ejecta, reaching over one half of the sky for the first few hours.

For the purposes of this initial exploration, we placed a uniform distribution of ^{56}Ni within the ejecta, enabling us to determine when the shock-heating effects diminish relative to the intrinsic light curve expected from the explosion. After about 12 hours, we find that the nickel

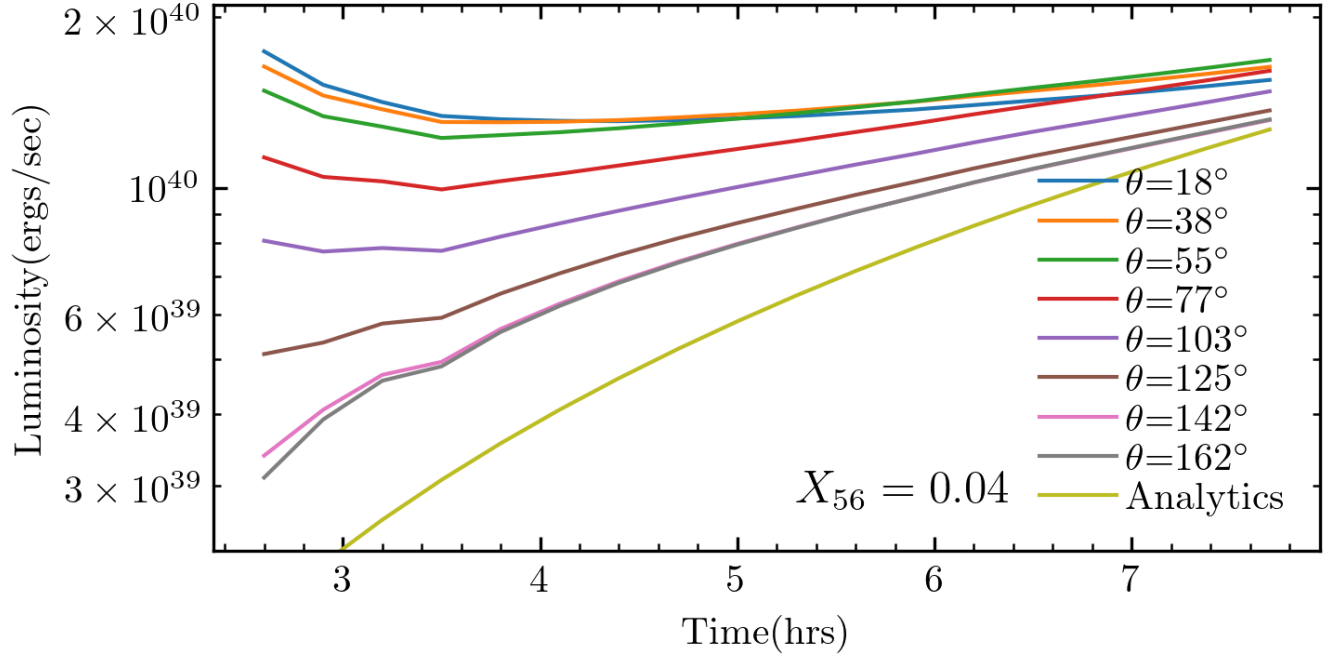


Figure 8. Luminosity vs time for $X_{56} = 0.04$ from $t \approx 2$ to 8 hrs. The legend provides the angle from which the observer views the supernova. We compare against the light curve predicted by Piro (2012) labeled “Analytics” describing an supernova without shock-heated ejecta.

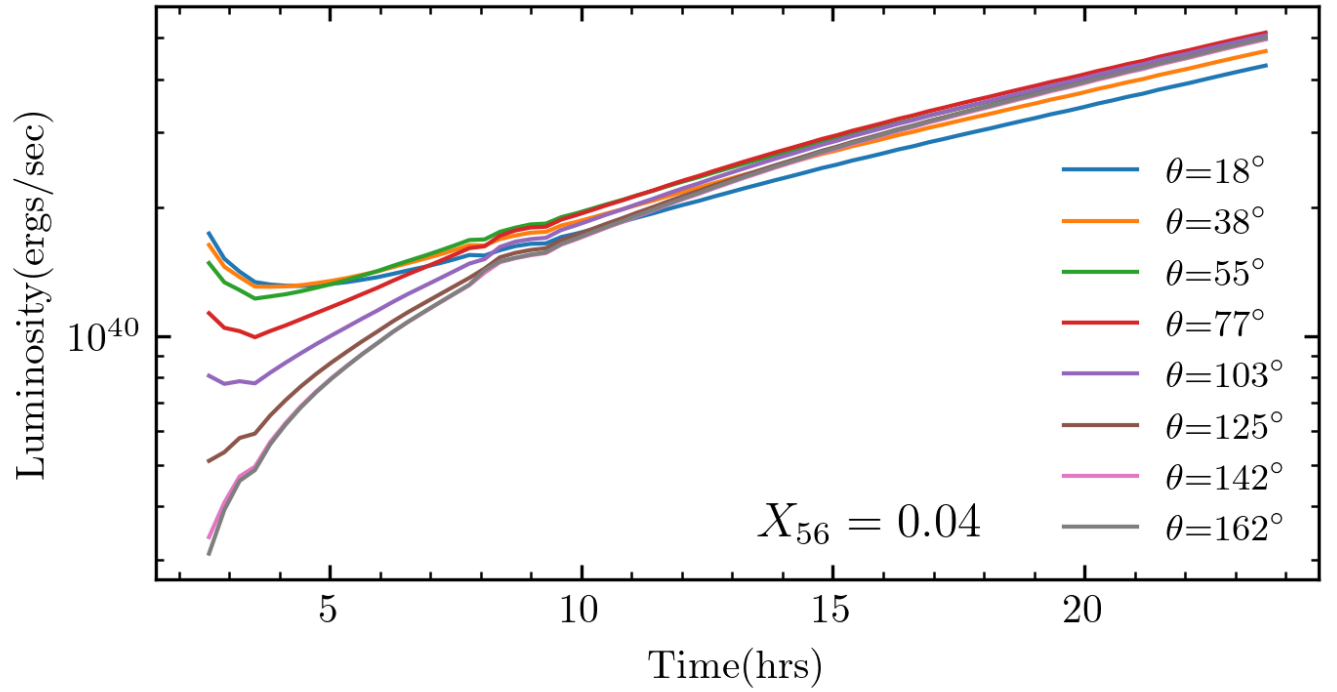


Figure 9. Luminosity vs time for $X_{56} = 0.04$ up to ≈ 1 day after the supernova at various viewing angles.

heating dominates the ejecta, so that most observers see nearly the same luminosity, which is in good agreement with the analytical predictions of Piro (2012). However, the modified density profile in the wake leads to a 15% underbrightness for observers directly in the wake compared to that seen in the unshocked ejecta. Whether this persists to later times near the peak of the light curve remains an open question.

Hence, there are two ways the effect of this collision can be seen. First, in the very early time light curves there will be an overbrightness which we calculate to be 1.5×10^{40} ergs/s at $t = 2.9$ hrs for the observer viewing from $\theta = 18^\circ$ as compared to the 3.9×10^{39} ergs/s that an observer viewing from the backside sees at that time for our assumed nickel mass. This overbrightness is roughly an order of magnitude smaller than that extrapolated from the work of Kasen (2010). This overbrightness decays away after about 12 hours. Second, at late times (i.e. one day) the interaction still has an effect, as observers at $\theta = 18^\circ$ then see 86% of the value seen by observers viewing the backside of the collision of 4.9×10^{40} ergs/s, and observers at $\theta = 38^\circ$ see 93%. This is due to the permanent marks on the ejecta density structure from the donor interaction. This may imply that the effects of the collision on creating the wake continue to impact the later light curves of Type Ia supernovae.

Though UV observations of SNIa have largely ruled out red giant or red supergiant donors (Brown et al. 2012), they allow for the possibility of binaries with smaller orbital separations corresponding to main-sequence or degenerate donors (Kutsuna & Shigeyama 2015). Alternatively, SNIa which show no sign of donor interaction may be merger products (Olling et al. 2015). However, the high-velocity ejecta found in this work may provide a mechanism for the two distinct ejecta velocities found in SN 2021aefx, which also contained the fastest ejecta observed in an SNIa to date (Ni et al. 2023). SN 2021aefx is a member of a subclass of SNIa which exhibit excess emission in the blue band over the first 0.5 to 2 days (Ni et al. 2024). Furthermore, Iskandar et al. (2025) conclude that in SN 2021hpr, early-excess emission is associated with high-velocity features in the Si and Ca lines. Hoogendam et al. (2025) found similar features in SN 2024epr, noting that neither a delayed detonation nor a thin shell He detonation sufficiently reproduced its observed properties. In future work, it will be necessary to perform multi-group radiative transfer calculations to obtain synthetic light curves which can be directly compared to those of the objects listed above.

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Software: Athena++ (Stone et al. 2020), Matplotlib (Hunter 2007), NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020)

APPENDIX

A. MOMENTUM TRANSFER EFFICIENCY

Hirai et al. (2018) computed the efficiency of momentum transfer from planar ejecta onto a spherical donor, assuming that the momentum $\mathbf{p}$ of an ejecta particle was deflected tangent to the surface of the donor at the point of intersection. Under these idealized assumptions they calculated that the fraction of incoming momentum imparted to the donor is $\eta_{\text{ideal}} = 1/2$, though in practice they found that η can vary from 0.17 to 0.72 depending on the hydrodynamic response of the donor. We extend this formalism to spherically-symmetric ejecta.

Consider a fluid parcel traveling outward from the supernova on a radial ray intersecting the spherical donor. The momentum imparted by the parcel when deflected through an angle σ is

$$p_{\text{imp}} = p \cos \zeta \sin \sigma, \quad (\text{A1})$$

where ζ is the angle at which the parcel intersects the donor surface as measured from the center of the donor. These angles are illustrated in Fig. 10 and are related by

$$\sin \sigma = \frac{\cos \zeta - R/a}{\sqrt{1 + (R/a)^2 - 2(R/a) \cos \zeta}} \quad (\text{A2})$$

for a donor with radius R and orbital separation a . The radial ray tangent to the donor will occur when $\sigma = 0$, corresponding to

$$\zeta_{\text{max}} = \pi/2 - \arctan(R/a). \quad (\text{A3})$$

Thus we need only consider the portion of the donor surface with $\zeta \leq \zeta_{\text{max}}$, as the remainder of the surface does not come into contact with ejecta in this formalism. The fraction of the total momentum of the ejecta intersecting the donor which is imparted to the donor is then

$$\eta_{\text{ideal}} = \frac{\int_0^{\zeta_{\text{max}}} p_{\text{imp}} \sin \zeta \cos \zeta d\zeta}{\int_0^{\zeta_{\text{max}}} p \sin \zeta \cos \zeta d\zeta}. \quad (\text{A4})$$

Eliminating σ and p_{imp} , this may be written as

$$\eta_{\text{ideal}} = \frac{\int_0^{\zeta_{\text{max}}} \sin \zeta \cos^2 \zeta \frac{\cos \zeta - R/a}{\sqrt{1 + (R/a)^2 - 2(R/a) \cos \zeta}} d\zeta}{\int_0^{\zeta_{\text{max}}} \sin \zeta \cos \zeta d\zeta} \quad (\text{A5})$$

Note that as $R/a \rightarrow 0$ this reduces to the case of planar ejecta

$$\lim_{R/a \rightarrow 0} \eta_{\text{ideal}} = \frac{\int_0^{\pi/2} \sin \zeta \cos^3 \zeta d\zeta}{\int_0^{\pi/2} \sin \zeta \cos \zeta d\zeta} = \frac{1}{2}. \quad (\text{A6})$$

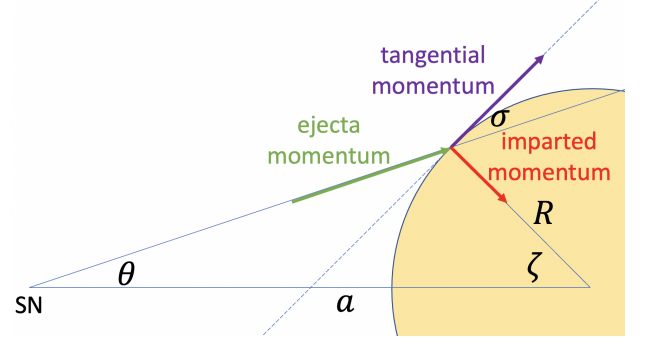


Figure 10. Illustration of spherically-symmetric ejecta impacting a spherical donor, with variable definitions. The ejecta momentum is decomposed into its components tangential and normal to the donor surface, the latter being imparted to the donor.

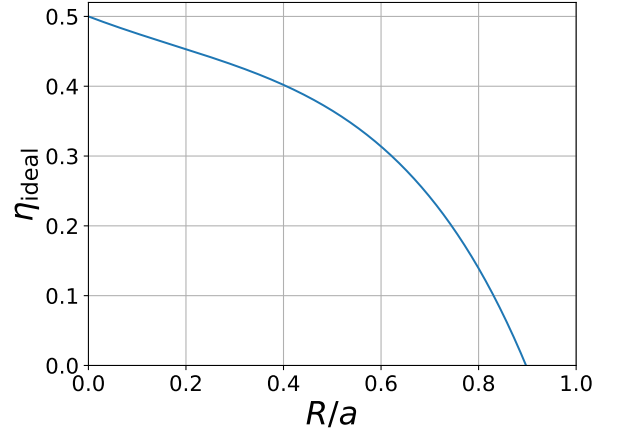


Figure 11. Ideal momentum transfer efficiency η_{ideal} versus R/a , based on (A5). For $R/a = 0$, (A5) reproduces the result for planar ejecta $\eta_{\text{ideal}} = 1/2$.

This is also the maximum possible value of η_{ideal} , as shown in Fig. 11, suggesting that momentum transfer is less efficient for spherical ejecta. The efficiency decreases monotonically with R/a but drops below zero only for clearly unphysical values of $R/a \gtrsim 0.9$. Because double-degenerate binaries have $0.3 \lesssim R/a \lesssim 0.4$ at Roche lobe overflow, η_{ideal} generally sits just above 0.4.

To measure this efficiency in our Athena++ simulation we must first compute the initial momentum of the ejecta

$$p_{\text{tot}} = \int_0^{v_{\text{max}}} 4\pi t^3 v^3 dv \rho(v). \quad (\text{A7})$$

Using the Gaussian ejecta profile presented in Wong et al. (2024) and truncating material with $v > v_{\text{max}}$,

this evaluates to

$$p_{\text{tot}} = \frac{2}{\sqrt{\pi}} M_{\text{ej}} v_0 \left[1 + \left(1 + \frac{v_{\text{max}}^2}{v_0^2} \right) \times \exp \left(-\frac{v_{\text{max}}^2}{v_0^2} \right) \right]. \quad (\text{A8})$$

Here $v_0 = \sqrt{(4/3)E_{\text{ej}}/M_{\text{ej}}}$ is the characteristic velocity. The fraction of the ejecta which intersects the donor is

$$\frac{\Omega}{4\pi} = \frac{1}{2} \int_0^{\arcsin(R/a)} \sin \theta d\theta. \quad (\text{A9})$$

We also need to project the momentum of this ejecta along the axis of symmetry, as transverse components cancel. This reduces the imparted momentum by a factor of

$$f_{\text{sym}} = \frac{\int_0^{2\pi} \int_0^{\arcsin(R/a)} \sin \theta \cos \theta d\theta d\phi}{\int_0^{2\pi} \int_0^{\arcsin(R/a)} \sin \theta d\theta d\phi}, \quad (\text{A10})$$

though this differs from unity by only a few percent. Then the total momentum of the ejecta imparted to the donor is $\Delta p = p_{\text{tot}} f_{\text{sym}} \Omega / 4\pi = 187 M_{\odot} \text{km/s}$ given the parameters of the explosion described in section 2.

Our simulation results give a momentum transfer integrated over the donor surface of $96 M_{\odot} \text{km/s}$, resulting in an efficiency of $\eta = 96/187 = 0.513$. This is $\approx 20\%$ greater than the value predicted by (A5) of $\eta_{\text{ideal}} = 0.431$. This excess in η/η_{ideal} roughly matches the stiffest stellar models of Hirai et al. (2018), who found $\eta \approx 0.6$ for many of their $N = 0$ and $N = 1.5$ polytropes. However, our treatment of the donor as a rigid body eliminates the stellar response of the donor to the collision as a complicating factor. Thus, the fact that we also find deviations from theory illustrates the limitations of the assumption that each gas parcel follows a ballistic trajectory, neglecting collisions within the fluid.

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